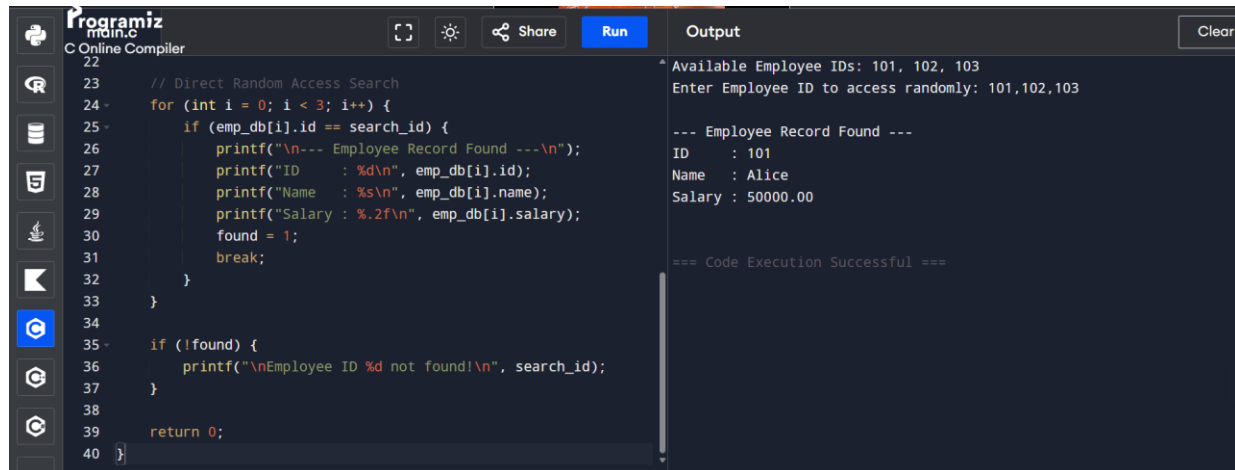


EXPERIMENT 16:

16. Develop a C program for implementing random access file for processing the employee details



The screenshot shows the Programiz Online C Compiler interface. The code editor on the left contains a C program for searching employee records. The output window on the right shows the program's execution results.

```
22
23 // Direct Random Access Search
24 for (int i = 0; i < 3; i++) {
25     if (emp_db[i].id == search_id) {
26         printf("\n--- Employee Record Found ---\n");
27         printf("ID      : %d\n", emp_db[i].id);
28         printf("Name     : %s\n", emp_db[i].name);
29         printf("Salary  : %.2f\n", emp_db[i].salary);
30         found = 1;
31         break;
32     }
33 }
34
35 if (!found) {
36     printf("\nEmployee ID %d not found!\n", search_id);
37 }
38
39 return 0;
40 }
```

Output:

```
Available Employee IDs: 101, 102, 103
Enter Employee ID to access randomly: 101,102,103

--- Employee Record Found ---
ID      : 101
Name     : Alice
Salary  : 50000.00

=== Code Execution Successful ===
```

PSEUDO CODE :

1. Define Employee struct (id, name, salary).
2. Open binary file using mode "rb+".
3. Write record: Seek end of file and write structure block.
4. Read record by ID: Seek byte offset $(id - 1) * \text{sizeof}(\text{Employee})$ using `fseek()`.
5. Read binary record directly using `fread()`.

CODE:

```
#include <stdio.h>

struct Employee {

    int id;

    char name[20];

    float salary;

};

int main() {

    FILE *fp = fopen("emp.dat", "wb+");

    struct Employee e1 = {1, "Alice", 50000.0};

    struct Employee e2 = {2, "Bob", 60000.0};

    struct Employee temp;

    // Write records sequentially

    fwrite(&e1, sizeof(struct Employee), 1, fp);

    fwrite(&e2, sizeof(struct Employee), 1, fp);

    // Random Access: Jump straight to second record

    int rec_id = 2;

    fseek(fp, (rec_id - 1) * sizeof(struct Employee), SEEK_SET);

    fread(&temp, sizeof(struct Employee), 1, fp);

    printf("Random Access Record %d:\nID: %d\nName: %s\nSalary: %.2f\n",

        rec_id, temp.id, temp.name, temp.salary);

    fclose(fp);
```

```
return 0;
```

```
}
```